

Analysis I: Proofs for Exam

Below, you'll find the list containing the 10 proofs we expect you to be familiar with for the examination. Please bear in mind that if a proof relies on another result, we don't envisage you to cite the result number, but rather provide a rephrased statement. For example, instead of stating 'By Theorem 2.57,' it is preferable to express something along the lines of 'In a previous lecture, we established the existence of the supremum for nonempty subsets that are bounded from above'.

1. **Convergent vs Cauchy Sequences (Theorem 2.124)**
 2. **Continuity = Sequential Continuity (Theorem 3.26)**
 3. **Intermediate Value Theorem (Theorem 3.29)**
 4. **Cauchy Root Criterion (Corollary 4.27)**
 5. **Chain Rule (Theorem 5.16)**
 6. **Rolle's and Mean Value Theorem (Theorems 5.30 & 5.31)**
 7. **Convexity vs Monotonicity of First Derivative (Proposition 5.51)**
 8. **Continuous Functions are Integrable (Theorem 6.35)**
 9. **Fundamental Theorem of Differential Calculus (Theorem 7.4)**
 10. **A Criterion for Analyticity at x_0 (Theorem 7.69)**
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